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EFFECTIVENESS OF AQUATIC LAND EXERCISE ON PHYSICAL PERFORMANCE AND PSYCHO-SOCIAL FACTORS AMONG INDIVIDUALS WITH KNEE OSTEOARTHRITIS

UPSI UNIVERSITI PENDIDIKAN SULTAN IDRIS

Kogilavani SKrishnan[1], Fairus Fariza Zainudin[1*], Mohansundar Sankaravel[1]

[1] Department of Health Science, Faculty of Sports Science and Coaching, Universiti Pendidikan Sultan Idris (UPSI) 35900, Tanjong Malim, Perak, Malaysia

INTRODUCTION

Knee Osteoarthritis (KOA) affects 20.5% of urban Malaysians. Over 71% of diagnosed KOA patients are overweight or obese, amplifying mechanical stress on lower extremity joints. Exercise is the core non-pharmacological treatment, yet adherence is low due to joint pain and kinesiophobia (fear of movement). Isolated land exercise can overload worn joints, while aquatic therapy lacks gravitational functional transfer.

OBJECTIVES

To evaluated the effectiveness of a structured aquatic-land exercise programme on physical performance and psychosocial outcomes among individuals with KOA.

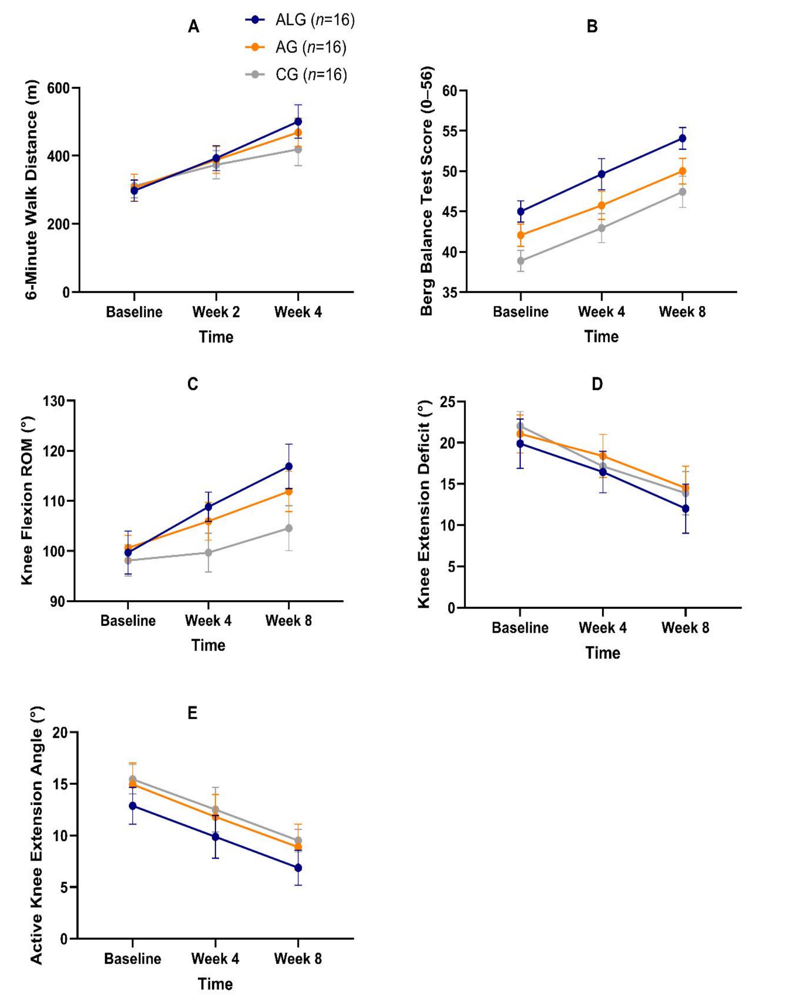
RESULTS

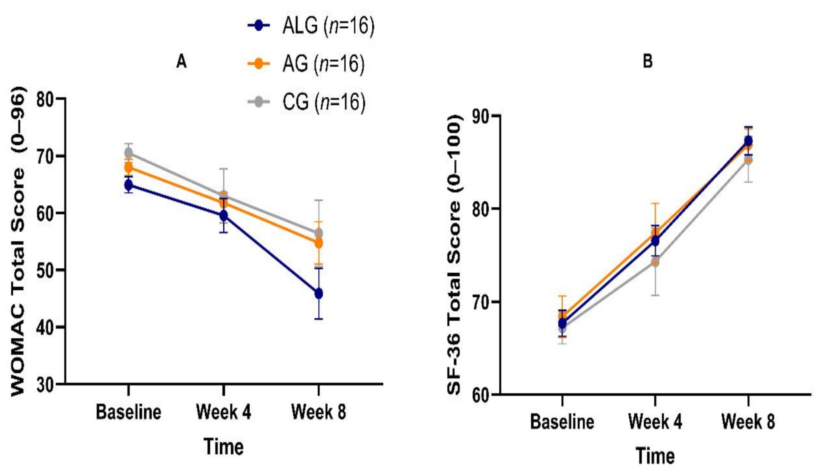
- After eight weeks, the ALG demonstrated significantly greater improvements in walking endurance, knee range of motion, balance, disability, and quality of life than the AG and CG ($p<0.001$–0.014).
- Combined aquatic-land exercise significantly outperformed aquatic-only and conservative care in enhancing walking endurance, balance, knee mobility, disability reduction, and quality of life, while psychological symptoms improved similarly across groups.
- Buoyancy unloads joint loads by (0% (ease pain & kinesiophobia), while land drills restore functional stability & weight bearing strength)

METHODOLOGY

- **Study setting**
  Physiotherapy Outpatient Department, Hospital Taiping, Perak
- **Data Collect**
  Data were collected at baseline, Week 4, and Week 8 for 48 participants measuring physical performance and psychosocial outcomes.
- **Statistical Analysis**
  Data normality was confirmed before analysis using One-way ANOVA & Mixed Repeated Measures ANOVA ($p<0.05$)
- **Age Range**
  45 - 65 years
- **Participants**
  48 participants with Knee osteoarthritis (14 male, 34 Female)allocated equally into 3 groups (n= 16 each).Baseline characteristics and outcome measures were comparable across groups
- **Intervention**
  The eight-week intervention compared three arms: combined aquatic-land exercise, aquatic exercise alone, and conservative care control, 3X weekly in 60-minute supervised sessions.

DATA VISUALIZATION





DISCUSSIONS

- Warm water buoyancy acts as an immediate pain modulator, reducing kinesiophobia and building movement confidence in a low-threat environment.
- Substantial disability reduction (WOMAC) and QoL enhancements require sustained 8-week exposure.
- Physical Gains (Week 4): Endurance and balance improvements manifest early in the initial 4-week window.

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CONCLUSION

The 8-week combined Aquatic-Land Exercise (ALE) programme produced superior physical performance (endurance, balance, range of motion), greater disability reduction, and higher quality of life compared to aquatic exercise alone or standard care in KOA patients.

RECOMMENDATIONS

- Integrate combined water and land based protocols into structured knee OA care guidelines within Malaysian public hospital physiotherapy departments.
- Conduct multi-center long-term follow-ups to evaluate retention effects beyond 8 weeks and evaluate cost-effectiveness across rural clinical settings.

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ETHICAL APPROVAL

National Medical Research Registry

- (NMRR): NMRR (24-00844-5YL(IIR); April, 2025)

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